Imagized Language Model: Structured Technical Documentation

Lachlan Chen AgInTiFlow
AgInTi Lab, LazyingArt LLC

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Abstract

The Imagized Language Model (ILM) treats language as an image-like tensor and generates text via diffusion-style denoising in that space. We present a structured specification with mathematical derivations for: superposition-based meta-elements, hierarchical memory-like embeddings, text rasterization, and discrete/latent diffusion. A practical roadmap targets consumer hardware.

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1 Executive Overview

ILM reframes text generation as image synthesis. A sentence becomes a compact, multi-channel 2D grid with explicit meta-element channels (grammar, semantics, tone, emotion). Generation proceeds coarse-to-fine via diffusion on discrete or continuous latents.

Key innovations

- Superposition of meta-elements (grammar, semantics, tone, emotion).
- Hierarchical mixed-radix vocabulary codes with spatial organization.
- Text-as-image rasterization and latent diffusion for efficiency.
- Multiscript support via glyph rendering and byte/character modeling.

2 Meta-Elements as Superposition

Let $\mathcal{G}=\{G_i\}$, $\mathcal{M}=\{M_j\}$, $\mathcal{E}=\{E_k\}$, $\mathcal{T}=\{T_\ell\}$ be basis sets for grammar, meaning, emotion, and tone. A token representation $s \in \mathbb{R}^d$ is modeled as

$$s \approx \sum_i \alpha_i G_i + \sum_j \beta_j M_j + \sum_k \gamma_k E_k + \sum_\ell \delta_\ell T_\ell + \varepsilon. \tag{1}$$

We learn dictionary D and sparse activations $\boldsymbol{a}$ via

$$\mathcal{L}_{\text{sup}} = \|s - D\boldsymbol{a}\|_2^2 + \lambda_1 \|\boldsymbol{a}\|_1 + \lambda_2 \sum_{u \neq v} \|D_u^\top D_v\|_F^2, \tag{2}$$

with the last term limiting cross-group interference. Disentanglement regularizers penalize $\mathrm{MI}(z_u, z_v)$ across factors.

Examples

- Grammar: UD relations (nsubj, obj, advmod), clause patterns (SVO/SOV/VSO), TAM, voice.
- Meaning: Frame primitives (CAUSE, GO, TO), role bindings (AGENT, THEME, RECIPIENT).
- Emotion/Style: VAD space and discrete categories; formality, politeness, certainty, register.

3 Hierarchical, Memory-Like Embeddings

Map $v \in \mathcal{V}$ to a mixed-radix code $\phi(v)=(k_1, \ldots, k_L)$ with $k_\ell \in \{0, \ldots, K_\ell - 1\}$ and capacity $\prod_\ell K_\ell \geq |\mathcal{V}|$. Arrange codes on a 2D grid (h, w) and learn $E[h, w] \in \mathbb{R}^d$.

Laplacian regularization With affinity W and degree matrix Deg, $L=\mathrm{Deg}-W$ induces

$$\mathcal{L}_{\text{spatial}} = \mathrm{tr}(E^\top L E) = \tfrac{1}{2} \sum_{p,q} w_{pq} \|E_p - E_q\|_2^2, \tag{3}$$

encouraging local smoothness aligned with lexical similarity.

Text-as-image rasterization Given width W, index i maps to (r, c) by $r=\lfloor (i-1)/W \rfloor$, $c=(i-1) \bmod W$. For level ℓ, form $X^{(\ell)} \in \{0, \ldots, K_\ell - 1\}^{H \times W}$; embed via $E_\ell \in \mathbb{R}^{K_\ell \times d_\ell}$, project to a common depth, and sum with meta-element channels and 2D positional encodings.

4 Diffusion for Language

Discrete diffusion Let $Q_t \in [0,1]^{K \times K}$ be the transition (e.g., absorbing-mask), and $\bar{Q}_t = \prod_{s \leq t} Q_s$. The single-site posterior is

$$q(x_{t-1} \mid x_t, x_0) \propto Q_t(x_{t-1} \to x_t)\, (\bar{Q}_{t-1})(x_0 \to x_{t-1}). \tag{4}$$

Training minimizes $\mathbb{E}_{t,x_0}\big[\lambda(t)\, \mathrm{KL}(q \| p_\theta)\big]$.

Latent diffusion Embed categorical planes to h_0; use Gaussian noising $q(h_t \mid h_0)=\mathcal{N}(\sqrt{\bar{\alpha}_t} h_0, (1-\bar{\alpha}_t) I)$. Operate in a compact latent z_t with an autoencoder and predict ε_θ.

Coarse-to-fine Assign per-level schedules $\beta_t^{(\ell)}$ so coarse levels retain information longer; choose schedules to maintain a target mutual-information proxy until step t^*.

5 Comparison with Autoregressive Generation

Aspect	Autoregressive	Diffusion (ILM)
Generation order	Sequential L→R	Parallel blocks, iterative refinement
Short outputs	Faster (< 50 tok)	Slower (multiple steps)
Long outputs	Slower (> 200 tok)	Faster (parallel denoise)
Error correction	One-shot, irreversible	Global revision each step
Controllability	Prompt heuristics	Guidance + meta channels
Perplexity	Lower (mature)	Approaching parity (2025)
Memory	KV cache grows	Fixed per step
Reasoning	Strong CoT	Emerging DoT
Creativity	Lower diversity	Higher diversity

Table 1: Autoregressive vs. diffusion generation.

6 Direct Character Visual Representation

ILM supports both byte/character encodings and glyph images. For CJK, render small glyphs (e.g., 32–64 px), pass through a tiny CNN, and fuse with text embeddings. For Latin scripts, ByT5-style byte models avoid OOV and improve spelling sensitivity.

7 Unified Objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{diff}} + \lambda_1 \mathcal{L}_{\text{sup}} + \lambda_2 \mathcal{L}_{\text{spatial}} + \lambda_3 \mathcal{L}_{\text{visual}} + \lambda_4 \mathcal{L}_{\text{recon}} + \lambda_5 \mathcal{L}_{\text{reg}}, \tag{5}$$

$$\mathcal{L}_{\text{reg}} = \lambda_{\text{ortho}} \sum_{u \neq v} \|D_u^\top D_v\|_F^2 + \lambda_{\text{dis}} \sum_{u \neq v} \text{MI}(z_u, z_v). \tag{6}$$

8 Architecture and Training

Backbone A UNet or Diffusion Transformer (DiT) operates on latent grids. Per-level embeddings are projected to a common depth and summed with meta-element conditioning and 2D positional encodings.

Sampling Use T=16–32 steps with classifier-free guidance. Coarse levels are generated first; fine levels condition on the skeleton.

Efficiency Latent diffusion yields $\times 16$–$\times 64$ fewer spatial elements; 8-bit quantization and LoRA reduce memory and adaptation cost.

9 Feasibility and Roadmap

We follow a phased plan: build hierarchical code tables; implement rasterization and latent AE; train a small UNet/DiT with discrete heads; add conditioning; then end-to-end finetune. Intrinsic metrics include NLL, edit distance, grammar accuracy; extrinsic tasks evaluate controllability and multilingual transfer.

Key insights Superposition enables control; hierarchical structure enables compression and interpretability; diffusion enables global revision; and visual encoding enables universal multilingual handling.

Appendix: Mathematical Notes

A. Graph Laplacian

With affinity W and degree matrix Deg, $L{=}\mathrm{Deg}{-}W$ yields $\mathrm{tr}(E^\top LE){=}\frac{1}{2}\sum_{p,q} w_{pq}\|E_p{-}E_q\|_2^2$.

B. Mixed-Radix Capacity

Given $C{=}\prod K_\ell$ codes and V items, balanced assignment reduces expected prefix collisions; nearest-neighbor lookup on prefixes accelerates decoding and improves robustness to denoise errors.